

MAJOR PROJECT - I | BACKEND DEVELOPMENT PROGRESS REPORT

Blockchain-Assisted Distributed AES Key Lifecycle Management Framework for Secure Healthcare Internet of Things

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1. Progress Summary

Since the last progress update, the team has completed the entire core AES key-lifecycle management engine — the security foundation of the project. All 14 planned backend phases covering encryption, key protection, full lifecycle management (rotation, expiration, archival, emergency recovery), authentication, authorization, audit logging, and transactional integrity are now complete, tested, and pushed to GitHub. This forms the complete Review 2 (Mid-Term) submission.

Overall Status: 14 / 33 total roadmap phases complete (Phases 1–14 — full backend security core) | Next Milestone: **Phase 15 — Validation (towards Review 3)**

2. Completed Backend Phases (1–14)

Phase	Module	Status	Key Work Completed
1–2	Foundation & Core Setup	● Completed	Spring Boot + Java layered architecture, Maven dependencies, MySQL/JPA connectivity.
3	Key Management (Base)	● Completed	AES-256 key generation, key metadata model, key APIs.
5–6	Security & Key Protection	● Completed	AES-256-GCM with random IV; master-key-based key protection layer.
7, 11–13	Full Key Lifecycle	● Completed	Create/Rotate/Revoke, scheduler-based Expiration, Archival, Emergency Recovery workflow (PENDING→APPROVED→COMPLETED).
8	Exception Architecture	● Completed	Centralized GlobalExceptionHandler, standardized error responses.
9	Auth & Authorization	● Completed	JWT login, BCrypt hashing, RBAC — verified 401/403/200 test matrix.
10	Audit Logging	● Completed	AuditLoggingFilter records actor, action, endpoint, timestamp, status.
14	DB Transactions	● Completed	@Transactional-backed atomicity — verified via key rotation DB test.

Verification: All modules independently tested (unit tests, Postman) with real evidence — see Review 2 (Mid-Term Experimental Result) document for detailed results. All code committed and pushed to GitHub.

3. Current Position — Backend Roadmap

The diagram below shows progress on the first 18 backend phases (full 33-phase roadmap covers blockchain and frontend in later reviews):



4. Next Milestone — Phase 15 Onward

Phase 15 — Validation

This phase focuses on strengthening input validation, edge-case handling, and result validation across the completed key-management engine — building toward Review 3 (Result Validation, 4th Week of November 2026). Phases 16–18 (Key Usage/Retrieval hardening, Advanced Rotation, API Documentation & Backend Hardening) follow to complete the backend before blockchain integration begins.

Roadmap Beyond Backend — Brief Overview

Stage	Phases	Focus
Before Review 3	15 – 18	Validation, secure key retrieval, advanced rotation, API docs (Swagger), testing, hardening — completes the backend.
Review 4 / 5 / 6 (Sem-7)	19 – 23	Blockchain fundamentals, architecture, smart contract, backend↔blockchain integration, verification.
Major Project II (Sem-8)	24 – 33	Full React.js frontend, user management, UI/UX polish, complete end-to-end integration.